

Abhay Sharma

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Research Interest

Passionate about Medical AI and Computer Vision with hands-on experience in medical imaging diagnostics, eye-monitoring hardware, and adversarial ML research. Seeking to apply deep learning and vision systems to build intelligent, accessible healthcare technology.

Education

Vellore Institute of Technology (VIT-AP University), Amaravati, India **2023 – 2027**
B.Tech in Computer Science and Engineering (Core) **CGPA: 9.40/10 (Top 10% of Branch)**

- GATE 2026 Qualified – Computer Science & Engineering, 2026
- Intermediate (Class XII) – CBSE, 2023

Technical Skills

Programming	Python, Java, JavaScript, SQL, C++ (Basics)
ML & Deep Learning	PyTorch, TensorFlow, Swin Transformer, ConvNeXt, YOLOv8, Scikit-learn
Medical AI	Medical Image Analysis, Multimodal Learning, Adversarial Robustness, Reinforcement Learning
Computer Vision	OpenCV, t-SNE, Latent Space Analysis, Image Segmentation, Object Detection
LLM & Agentic AI	CrewAI, MCP, RAG Pipelines, Prompt Engineering, LangChain (Basics)
Frameworks & Tools	FastAPI, Spring Boot, React.js, Flask, Node.js, Express.js
Databases	MongoDB, Neo4j (Graph DB), SQL, FAISS
Hardware	Arduino Uno, Embedded Systems, Sensor Integration
Cloud & DevOps	AWS, Kafka (Basics – Event Streaming), Docker, Git, Linux
Core CS	OS, CN, DBMS, OOPs, DSA, Software Engineering

Projects

Multiview Learning for Skin Lesion Detection **Medical AI – 2025**

- Developed a multiview deep learning pipeline fusing dermoscopic images with patient metadata for skin lesion classification on the HAM10000 dataset.
- Achieved 94% accuracy using an ensemble of ConvNeXt CNN and Swin Transformer architectures with multimodal feature fusion.
- Implemented multimodal fusion strategy combining visual features with structured metadata (age, lesion location, sex) for superior diagnostic precision over single-modality models.
- Directly applicable to medical imaging diagnostics in ophthalmology, retinal screening, and clinical AI systems.

Adversarial World Model Security – AdvWM Research Project **2025 – 2026**

- Pioneered Imagination Poisoning attacks on DreamerV3 world models in model-based RL, achieving 87.2% Attack Success Rate with stealthiness score of 0.037 KL divergence.
- Implemented 4 novel attack vectors (Gradient Latent Attack, GAN Poisoner, Replay Poisoning, Temporal Attack) and 4 defense mechanisms including certified robustness with L2 radius $R=0.459$.
- Built real-time research dashboard using React + D3.js with live t-SNE latent space visualization and WebSocket streaming via Spring Boot API gateway.
- Designed 3-tier architecture: PyTorch ML engine → Spring Boot proxy → React frontend; all 7 research metrics meeting NeurIPS/ICML 2026 submission targets.

BlinkEye – Multivision Driver Alert System **Hardware – 2024**

- Designed and built an eye-blink monitoring safety system using Arduino Uno and infrared eye sensors to detect drowsy driving in real time.
- System triggers an audio alert when eyes remain closed for more than 3 seconds, preventing sleep-induced road accidents.
- Demonstrated end-to-end hardware integration: sensor input → microcontroller logic → alert output, directly relevant to embedded medical device development.

AI-Based Smart Surveillance & Public Safety Platform **2025**

- Developed a real-time surveillance system using YOLOv8 for detecting public misconduct with automated alert generation.
- Designed REST APIs for video stream processing and integrated IoT-based alert mechanisms for real-world deployment.

- Enterprise Knowledge Assistant using LLMs

2025
- Built an LLM-powered assistant using Retrieval-Augmented Generation (RAG) for document-based Q&A with semantic search.
 - Designed Flask-based APIs with modular architecture; focused on prompt optimization to minimize hallucinations.

Experience & Leadership

- Campus Mantri (Official Representative) – GeeksforGeeks

2026 – Present
- Acted as the official liaison between GeeksforGeeks and VIT-AP University.
 - Led coding initiatives, workshops, and competitive programming events for 200+ students.
- President – GeeksforGeeks Student Chapter, VIT-AP

2026 – Present
- Founded and led the official GeeksforGeeks student chapter; managed speaker sessions and technical mentorship programs.
- Technical Lead – Microsoft Student Chapter (MSC-VITAP)

2025 – Present
- Led technical development for AI-based and cloud-focused projects; guided members on Microsoft technologies.
- Lead, AI/ML Team – Android Club, VIT-AP

2025 – Present
- Mentored 10+ members in ML pipelines, project architecture, and deployment workflows.

Certifications

- Oracle Certified Generative AI Professional
- Oracle Certified Data Science Professional